The Great Pumpkin Snatch

Whodunnit? · Exponents Name: _____

It was the day of the Harvest Fair, and the biggest pumpkin in county history sat on its pedestal. Then the judges came to weigh it. The pedestal was empty: **Big Orange**, the 400-pound champion pumpkin, was gone!

Six people were at the fair, each in a different place, and there are six ways to move a giant pumpkin. Put on your detective hat: **who** took it, **where** were they, and **how** did they do it?

How to crack the case

1. Your teacher will post **10 clue cards** around the room. Each card has a math problem and four answer choices. Every choice is paired with a clue.
2. Solve the problem on the **Show Your Work** page. Find your answer among the choices and read the clue next to it. If your answer is not there, check your work.
3. Use the clue to cross out suspects, places or methods on the case board below. Write the card number beside what it cleared.
4. After all 10 cards, exactly one suspect, one place and one method are left. Those three boxes name the culprit. If you end up with none, or more than one, a clue went wrong: recheck your math!

The Suspects	The Places	The Methods
☐ Farmer Whit Pumpkin Grower	☐ Pumpkin Patch	☐ Rolled downhill
☐ Mrs. Alder Fair Organizer	☐ Hay Maze	☐ Loaded on a hay wagon
☐ Mr. Corbin Hayride Driver	☐ Cider Barn	☐ Hidden under a tarp
☐ Dottie Pike Pie Judge	☐ Judging Stage	☐ Hauled by tractor
☐ Old Pete Scarecrow Maker	☐ Petting Zoo	☐ Carted in a wheelbarrow
☐ Ms. Hale Cider Seller	☐ Parking Field	☐ Dressed as a scarecrow

The culprit is _____ in _____ using _____

Show Your Work

The Great Pumpkin Snatch · Show every step in the matching box.

Name: _____

1. Choice _____	**2.** Choice _____
3. Choice _____	**4.** Choice _____
5. Choice _____	**6.** Choice _____
7. Choice _____	**8.** Choice _____
9. Choice _____	**10.** Choice _____

Clue #1

The Great Pumpkin Snatch

Simplify:

$$x^4 \cdot x^3$$

A. x^7	Farmer Whit was weeding the Pumpkin Patch. Three neighbors waved to him.
B. x	Farmer Whit was directing trucks in the Parking Field. His orange vest was seen by a dozen drivers.
C. x^{12}	Mr. Corbin was sipping hot cider in the Cider Barn. The cashier remembers his big tip.
D. $2x^7$	Every tarp at the fair was lifted and checked. Each one covered only folding chairs.

Solve the problem, find your answer, and read its clue. Cross it off on your case board.

✂ cut here

Clue #2

The Great Pumpkin Snatch

Simplify:

$$\frac{x^8}{x^3}$$

A. x^{-5}	The blue tarp by the barn is soaked and wrinkled flat. Nothing big has been under it.
B. x^{24}	Ms. Hale was selling cider from a cart in the Parking Field. Her sales slip shows the time.
C. x^{11}	Mr. Corbin was feeding carrots to the goats in the Petting Zoo. A little girl took his picture.
D. x^5	The hill behind the patch is untouched. No pumpkin rolled down it.

Solve the problem, find your answer, and read its clue. Cross it off on your case board.

Clue #3

The Great Pumpkin Snatch

Simplify:

$(x^3)^4$

A. x^{81}	Dottie Pike was handing out pie samples in the Parking Field. The line of hungry visitors vouches for her.
B. x^{12}	Mrs. Alder was announcing winners from the Judging Stage on the loudspeaker.
C. x^7	Mr. Corbin was posing with a giant squash in the Pumpkin Patch. The fair photographer has the shot.
D. x	The tarp from the tool shed was folded neatly with its corners tucked. It hasn't been moved.

Solve the problem, find your answer, and read its clue. Cross it off on your case board.

✂ cut here

Clue #4

The Great Pumpkin Snatch

Simplify:

$(2x^3)^2$

A. $4x^6$	The hay wagon was empty. It only held bales all day.
B. $2x^6$	Mr. Corbin was clapping for the winners at the Judging Stage. The front row saw him the whole time.
C. $4x^9$	Tarp inspectors found no lumps, bumps or pumpkin-shaped shapes anywhere on the grounds.
D. $4x^5$	Old Pete was fixing a flat tire in the Parking Field. The tire shop's receipt has the time.

Solve the problem, find your answer, and read its clue. Cross it off on your case board.

Clue #5

The Great Pumpkin Snatch

Simplify:

$$3x^2 \cdot 4x^3$$

A. $12x^5$	Dottie Pike was tasting pumpkin pies in the Cider Barn. The judge’s card has the time.
B. $12x^6$	Mr. Corbin was helping lost kids out of the Hay Maze. The kids still talk about him.
C. $7x^5$	Mrs. Alder was handing out ribbons in the Parking Field. The ribbon box count backs her up.
D. $12x$	The big green tarp has no pumpkin stains on it. It only ever covered the hay bales.

Solve the problem, find your answer, and read its clue. Cross it off on your case board.

✂ cut here

Clue #6

The Great Pumpkin Snatch

Simplify:

$$\frac{18x^7}{6x^2}$$

A. $3x^9$	Mr. Corbin was refilling the cider jugs in the Cider Barn. The jug count matches the schedule.
B. $3x^5$	The tractor never started. Its engine log shows it parked all evening.
C. $3x^{14}$	The tarp over the tractor was lifted for the inspector. Only an old toolbox was underneath.
D. $12x^5$	Old Pete was patching a scarecrow in the Parking Field. Stray straw on the ground proves it.

Solve the problem, find your answer, and read its clue. Cross it off on your case board.

Clue #7

The Great Pumpkin Snatch

Write with a positive exponent:

$$x^{-3}$$

A. x^3	Mr. Corbin was tying ribbons on vines in the Pumpkin Patch. The ribbon spool is almost empty.
B. $-\frac{1}{x^3}$	A search of the tarps found nothing under them but puddles and a lost mitten.
C. $-x^3$	Farmer Whit was handing out maps in the Parking Field. The gate attendant wrote down the time.
D. $\frac{1}{x^3}$	Old Pete was stuffing scarecrows in the Hay Maze with a group of kids.

Solve the problem, find your answer, and read its clue. Cross it off on your case board.

✂ cut here

Clue #8

The Great Pumpkin Snatch

Simplify:

$$5x^0 \quad (x \neq 0)$$

A. -5	The tarp rope knots are still tied the way the crew left them. Nobody has untied them.
B. 1	Mrs. Alder was greeting visitors in the Parking Field. The guest book has her handwriting.
C. 0	Mr. Corbin was braiding a pony's mane in the Petting Zoo. The zoo keeper signed him in.
D. 5	The wheelbarrow axle is fine and not bent. It never held 400 pounds.

Solve the problem, find your answer, and read its clue. Cross it off on your case board.

Clue #9

The Great Pumpkin Snatch

Solve for x:

$$3^x = 81$$

A. x = 9	Ms. Hale was loading apple crates into a truck in the Parking Field. The driver kept her receipt.
B. x = 3	The lumpy tarp turned out to cover a pile of firewood.
C. x = 4	Ms. Hale was selling cider at the Petting Zoo. The goats got the leftovers.
D. x = 27	Mr. Corbin was fixing a speaker cable at the Judging Stage. The sound crew watched him work.

Solve the problem, find your answer, and read its clue. Cross it off on your case board.

✂ cut here

Clue #10

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An account starts at $500 and grows 20% each year. Write the exponential function for the balance y after x years.

A. $y = 500(1.2x)$	Dottie Pike was tasting tailgate brownies in the Parking Field. Her scorecard has the time on it.
B. $y = 500(20)^x$	Not one tarp on the fairgrounds has been moved since morning. The dew on them is undisturbed.
C. $y = 500(1.2)^x$	The scarecrows at the fair all have straw bellies. None hides a pumpkin.
D. $y = 500(0.2)^x$	Mr. Corbin was untangling a hay bale pile in the Hay Maze. Three kids cheered him on.

Solve the problem, find your answer, and read its clue. Cross it off on your case board.